TransLab / Tantor AI Platform – Product Requirements Document

1. Product Overview

The product is an enterprise-grade AI and data platform (referred to in the transcript as “Tantor / Tanto AI workload” and an "agentic AI platform"), designed primarily for banks and financial institutions to build, deploy, and govern AI/ML and agentic workflows such as credit decisioning, cross-sell/upsell, churn analytics, multimodal RAG, and internal knowledge assistants.[^1]

The platform unifies data ingestion, data federation/virtualization, data governance, AutoML, explainable ML (XAI), and agentic workflow orchestration into a single system, with drag-and-drop configuration aimed at reducing the need for deep coding skills while still supporting advanced ML operations.[^1]

Primary flagship use cases highlighted in the transcript:

- Credit underwriting and agentic credit decisioning with human-in-the-loop.
- Cross-sell and up-sell model building for banking products.
- Churn analytics, customer 360, customer segmentation.
- Multimodal RAG over regulatory documents and knowledge artefacts.
- Governance and observability of AI models and agentic workflows.[^1]

2. Goals and Non-Goals

2.1 Product Goals

- Enable banks to quickly integrate disparate core and peripheral systems (on-prem and cloud) into a unified data and AI platform via prebuilt connectors and data ingestion pipelines.[^1]
- Provide a governed data layer (federated and/or physical) that supports analytics, AI/ML training, and consumption via APIs without always requiring a data warehouse build.[^1]
- Offer a full AutoML / ML workflow builder (data profiling, cleaning, feature engineering, model training, XAI, fairness checks, deployment, monitoring) through a low-code interface.
- Deliver an agentic AI credit decisioning framework that is explainable, auditable, and suitable for regulated financial environments, with human-in-the-loop controls and detailed execution traces.[^1]
- Provide strong AI governance (fairness checks, data/model drift monitoring, guardrails) and operational observability at both workflow and agent/model level.

- Reduce time-to-market for new AI use cases (credit underwriting, cross-sell, churn, multimodal RAG, etc.) through prebuilt blueprints/use cases and re-usable modular agents.[^1]

2.2 Non-Goals

- The platform does not aim to be a core banking system; it integrates with core and peripheral systems.
- It does not replace bank-specific risk policy engines but should integrate with them or encode their policies via configurations and rule layers.
- It does not attempt to fully solve all regulatory fairness and FEAT / FREE-AI style requirements in the first version; it provides hooks (fairness attributes, governance reports, XAI) and evolves with client feedback.[^1]

3. Target Users and Personas

- **Data Scientists / ML Engineers**: Want an integrated environment for data access, profiling, feature engineering, model training, and XAI without juggling multiple tools.
- **Quantitative Risk & Credit Teams**: Need configurable but governed credit decisioning pipelines with visibility into model behavior, risk factors, and audit trails.
- **Business Product Owners (Loans, Cards, Retail Banking)**: Need faster launch of AI-driven products (STP/partial STP loan flows, cross-sell journeys) without deep technical intervention.
- **Data Engineers / Platform Teams**: Need robust ingestion, federation, and governance capabilities with connectors into existing data estates.
- **Compliance & Model Risk Management**: Require fairness checks, governance reports, explainability artefacts, logs, and overrides with justification.
- **Executives**: Need dashboards on AI performance, automation coverage, portfolio risk, token/LLM costs, and workflow KPIs.[^1]

4. High-Level Architecture & Modules

The platform is conceptually split into three vertical layers plus agentic orchestration and governance overlays.[^1]

4.1 Source & Ingestion Layer

Core responsibilities:

- Connect to 50+ prebuilt source systems including SQL/NoSQL databases, enterprise and cloud databases, applications, file-based sources, and object stores like S3 and iODS.[^1]
- Provide UI-driven connector setup (host, port, credentials) and persist connection objects.
- Support batch and streaming ingestion modes based on customer needs.

- Offer an in-house ETL/ELT tool to build data warehouses or data lakehouses on-prem or on cloud.

Key functional requirements:

- Connector catalogue with metadata (type, required parameters, capabilities).
- Secure credential management (vault, rotation, RBAC-based access to connectors).
- Data ingestion jobs with schedule, source, transforms, target, monitoring.
- Error handling, retries, failure notifications.

4.2 Data Federation & Virtualization Layer

Core responsibilities:

- Allow creation of virtual views over distributed source systems without physically moving data.[^1]
- Support joining tables across multiple systems and exposing them as queryable datasets.
- Enable exporting of federated data, training models directly on federated views, and exposing federated views via APIs for downstream consumption.[^1]

Key functional requirements:

- UI to define virtual datasets (sources, joins, filters, projections).
- Pagination support and performance optimization (e.g., pushing down predicates, caching, materialized views).[^1]
- Ability to materialize frequently used virtual views for performance, with refresh scheduling.
- Access control and lineage tracking for virtual datasets.

4.3 Data Governance Layer

Core responsibilities:

- Data security and privacy: masking, hashing, access controls for PII and sensitive fields.[^1]
- Role-based access control (RBAC) for datasets, connectors, models, and workflows.
- Data quality rules (completeness, validity, uniqueness, etc.) with scoring and alerts.[^1]
- Data lineage: track data origin, transformations, and downstream usage.
- Data catalog: searchable inventory of datasets with metadata, ownership, and classifications.[^1]

Key functional requirements:

- Governance rule engine to define and apply rules across datasets.
- Data quality dashboards and reports.
- PII detection and classification, with policy-driven masking.

- Integration with external IAM/IDP systems (e.g., AD, LDAP, SSO).

4.4 Consumption & AI/ML Layer

Core responsibilities:

- Provide an AI Playground where users can create projects, add datasets (from physical or federated layers), and run profiling, model building, and deployment flows.[^1]
- End-to-end AutoML and manual ML pipelines:
  - Data profiling report (overview, alerts, reproduction details).[^1]
  - Data cleaning pipeline with ~7 steps (missing value handling, outlier detection/treatment, encoding, transformations, variance filtering, scaling, etc.).[^1]
  - Support for 4 prediction types and multiple algorithms (binary, multi-class, regression, 40+ algorithms grouped by family).[^1]
  - Train-test split configuration (default 90/10) with validation that split sums to 100.[^1]
  - Manual and auto modes for data preprocessing and model selection.
  - Hyperparameter tuning modes (basic/default, hyperparameter mode, grid search).[^1]
- Model evaluation & comparison dashboards (accuracy, precision, confusion matrices, etc.).[^1]
- XAI reports using multiple techniques (SHAP, LIME-like methods, partial dependence plots, counterfactual explanations, feature importance and flip analysis).[^1]
- Model saving, versioning, and deployment to inference endpoints.

Key functional requirements:

- Project and dataset management (ownership, sharing, access control).
- Step-wise pipeline builder with ability to go back to specific steps and reprocess data.[^1]
- Support for notebook upload and custom model integration.
- Export evaluation and XAI artefacts (PDF/CSV) for governance.

4.5 Scheduling, Drift & Orchestration

Core responsibilities:

- Model orchestration engine to schedule inference runs with configuration of project, model, data source, destination tables, and validation steps.[^1]
- Guardrails for data and model drift with thresholds per deployment; if breached, trigger alerts and optionally block predictions.[^1]
- Support separate flows for batch and (future) real-time scoring.

Key functional requirements:

- Schedule types (cron-like, fixed date/time, timezone aware).
- Drift metrics and dashboards.

- Alerting channels (email, future webhook integrations) when thresholds are breached.[^1]

4.6 Agentic AI Studio & Credit Decisioning Framework

Core responsibilities:

- Agent Studio for composing modular AI agents into structured, reusable workflows.[^1]
- Support for structured and unstructured inputs (JSON, documents) via data nodes and classification agents.[^1]
- Specialised sub-agents for document verification, income and bank statement analysis, risk assessment (financial, behavioural, documentation), and collateral checks.[^1]
- Orchestration agent that consolidates agent outputs, applies business rules and AI reasoning, and produces a final decision (e.g., approve/reject) with confidence scores and rationale.[^1]
- Decision cards UI showing status, key risk drivers, pros/cons, and execution trace per application.[^1]
- Maker-checker flows with human-in-the-loop approvals, overrides with justification, and immutable audit trail of changes (timestamp, user, old/new decisions, reason).[^1]

Key functional requirements:

- Drag-and-drop agent workflow builder with configurable inputs, outputs, and policies per agent.
- Ability to parameterize flows per product (e.g., home loan vs. two-wheeler vs. SME) and segment.
- Integration hooks to existing LOS/LMS so platform can work as a decisioning/service layer rather than requiring end-to-end LOS replacement.[^1]
- Configurable credit policies and score cut-offs to drive decision logic.

5. Detailed Feature Requirements

5.1 Data Connectors & Ingestion

- Maintain catalogue of 50+ connector types with an extensibility framework for new connectors.[^1]
- Connection test feature to validate connectivity before saving.
- Support for incremental loads and full loads.
- Support for secure file transfer (SFTP, HTTPS) and object store access.
- Logging of ingestion job runs with metrics (rows processed, errors, duration).

5.2 Data Federation & Virtualization

- Visual query builder to define joins across systems without SQL.
- Advanced users can switch to SQL mode.
- Pagination and result size limits with offset/page tokens to protect performance.[^1]
- Materialized view management (create, refresh schedule, on-demand refresh, status).

5.3 Data Governance

- Data masking templates (e.g., mask PAN, Aadhaar, phone, account numbers) configurable by policy.
- Hashing options for irreversible pseudonymization.
- Data quality rules library and rule builder; attach rules to datasets and pipelines.[^1]
- Quality score computation and visualization per dataset.
- Lineage view showing end-to-end flow from source to models and reports.

5.4 AI Playground & AutoML

- Project creation UI: project name, description, team members, datasets.[^1]
- Dataset selection from federated or physical tables with optional filtering.
- Automated data profiling resulting in three main sections: overview, alerts, reproduction/report tabs.[^1]
- Alerts for high correlation, missing values, skewness, etc.[^1]
- Column-wise drill-down: statistics, histograms, frequent/extreme values, word clouds.[^1]
- Seven-step data cleaning pipeline components (toggleable per step) including:
  - Unique ratio checks and duplicate removal.
  - Low-information column removal.
  - Missing value strategies (drop, numeric imputation, categorical imputation) with per-column overrides.[^1]
  - Outlier detection and treatment strategies.
  - Encoding of categorical variables.
  - Transformations for skewed data (e.g., log, Box-Cox) and scaling (7+ scaler methods).[^1]
  - Near-zero variance removal.
- Prediction type selection: classification (binary, multiclass) and regression; supervised learning workflow.[^1]
- Model training support for 40+ algorithms grouped into linear, kernel, tree-based, ensemble, etc.[^1]
- Model configuration modes:

    - Baseline (default params).
    - Hyperparameter tuning.
    - Grid search.
- Model comparison charts and tables (accuracy, precision, recall, F1, ROC AUC, etc.).[^1]

5.5 Explainable AI & Fairness

- XAI suite with at least five techniques (e.g., SHAP, LIME-like methods, PDP, counterfactual explanations, feature importance/flip analysis).[^1]
- Feature importance display per model and per prediction.
- Counterfactual reports showing minimal changes required to flip a decision.[^1]
- Live per-prediction explainability view.
- Fairness attributes configuration (protected attributes like gender, age bands, etc.) per model.[^1]
- Fairness assessment algorithms that classify model as fair/unfair based on configured metrics (demographic parity, equal opportunity, etc.) with clarity in the governance report.[^1]
- Governance report download (PDF/CSV) at model save time, containing performance, fairness metrics, and XAI summary.[^1]

5.6 Drift, Guardrails, and Monitoring

- Data drift and model drift detection per deployment, with configurable thresholds.[^1]
- When thresholds are breached:
    - Send alert via configured channels.
    - Optionally block predictions until data/model is remediated.[^1]
- Support offline “what-if” retraining or data extension when drift is detected.
- Monitoring dashboards at:
    - Workflow level: success rate, end-to-end latency (e.g., P95), decision acceptance rate, automation coverage, override frequency, incident rate.[^1]
    - Agent level: availability, task success rate, output consistency, safety compliance flags, latency, token usage and cost per execution.[^1]
- Export monitoring and governance reports as CSV/PDF for board and regulator reporting.[^1]

5.7 Agentic Credit Decisioning

- Prebuilt credit underwriting workflow template:
    - Input node: accept LOS payload (structured) and document set (unstructured).
    - Document verification agent (including optional checksum/file integrity validation hooks).[^1]

    - Income and bank statement analysis agents.
    - Risk and collateral assessment agents for financial and behavioural risk.
    - Orchestration agent applying rules and AI reasoning.
    - Output node producing decision + confidence + reasoning + risk factors.
- Maker-checker flow:
    - Role definitions (maker, checker, admin).
    - Ability for maker to accept/override AI suggestion with justification.
    - Checker to validate and finalize.
    - Immutable audit log per application.
- Policy configuration:
    - Define segments, products, thresholds and special rules.
    - Map products and segments to different workflows or agent configurations.
- STP vs. non-STP configuration:
    - For standard, low-risk products, allow STP with thresholds and optional random sampling for manual review.
    - For complex cases, enforce human-in-the-loop.

5.8 Multimodal RAG & Knowledge Assistants

- Ability to ingest documents including PDFs, scanned images, and potentially screenshots (using OCR and LLMs) to build a knowledge base.[^1]
- Support multimodal RAG over regulatory documents, product manuals, and internal process documents.[^1]
- Chat-style interface or API for question-answering on top of this knowledge base.
- Use cases:
    - Regulatory query assistant (e.g., RBI circulars).[^1]
    - Support for internal knowledge artefacts without pre-existing metadata.

5.9 Executive & Operational Dashboards

- Executive dashboard:
    - Total applications processed.
    - Approval/rejection rates.
    - Portfolio risk exposure.
    - AI metrics including token usage, system accuracy.
- Operational views:
    - Per workflow and per agent metrics as described in monitoring section.

6. Constraints, Risks, and Open Questions

6.1 Known Constraints

- Some capabilities (e.g., agricultural collateral verification, domain-specific technical valuation of plant and machinery) are explicitly called out as not yet built and are considered roadmap items.[^1]
- Current fairness handling often expects the customer to adjust data (e.g., add more diverse training data or segment data) rather than automatically re-balancing or transforming for fairness; this raised regulator-alignment concerns in the discussion.[^1]
- Some demonstrations in the transcript used synthetic banking data; production deployments must be retrained on actual bank data per institution.[^1]

6.2 Risks

- Regulatory risk if fairness and FEAT / FREE-AI principles are not deeply integrated and auditable; splitting models by gender or similar protected attributes is viewed as non-compliant.[^1]
- Over-promising multimodal and collateral verification capabilities before they are fully production-ready across languages and geographies (e.g., India's agricultural land records in multiple languages).[^1]
- Complexity and learning curve if the platform becomes too feature-heavy; mitigated by drag-and-drop design and 1–2 day enablement workshops.[^1]

6.3 Open Questions & To-Be-Defined

- How to encode and manage complex bank-specific credit policies at scale (per segment, product, geography) while maintaining maintainability and transparency.
- Detailed design for fairness remediation flows (not just assessment) that align with RBI/MAS guidelines without requiring customers to create prohibited segmented models.
- Extent of support for scanned documents and low-quality images in multimodal RAG pipelines and how to robustly handle them.
- Strategy for out-of-the-box STP vs. advisory recommendations for human-in-the-loop in different product lines.
- Licensing & deployment models: fully managed SaaS vs. bank-hosted vs. hyperscaler marketplace offerings.

7. Success Metrics

- Reduction in model development time for a new use case (from weeks/months to days).
- Reduction in credit decision turnaround time while maintaining or improving portfolio risk metrics.
- Percentage of decisions executed via governed agentic workflows vs. manual-only processes.

- Fairness and governance adoption: number of models with documented fairness assessments and governance reports.
- Platform adoption metrics: number of active projects, workflows, and agents; number of bank clients onboarded; number of prebuilt use cases reused.
- User satisfaction (NPS) for data scientists, risk teams, and business users.

8. Roadmap Themes (Indicative)

- **Regulatory-grade Governance Enhancements**: Deep integration with FEAT/FREE-AI style guidelines, more robust fairness remediation tools, and regulator-ready templates.
- **Domain Packs for Credit**: Pre-configured blueprints for various loan products (retail mortgage, auto, two-wheeler, SME, agri) including domain features, policy templates, and sample workflows.
- **Advanced Multimodal**: Stronger support for diverse document formats, low-resource languages, and computer-vision-based collateral assessment.
- **Ecosystem & Marketplace**: Hyperscaler marketplace listings (AWS, Azure), partner integrations, and templates for consulting partners.
- **STP Optimisation**: Enhanced STP capabilities for standard products with fine-grained risk guardrails and post-facto sampling.

References

1. Voice-260429_143419-TransLab_original.txt -
   Speaker 1 (00:39)
   Manish, for our future reference, can we record this call? Is it fine?

Speake...